IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

# Charging Infrastructure

## Lecture-57

## DC Charging System(BEVC-DC001)-III

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# Recap

- DC001 DC charging system
- CAN Communication

# Recap: CAN Database File

- These are the text files containing information of CAN messages, signals and their parameters
- Each CAN message can have multiple CAN signals.
- The device manufacturers usually provide the CAN database file along with their product. For each signal, CAN databases define the rules for conversion to engineering units.
- The following data is stored in databases:
  - Message name, Message ID and DLC bit
  - Channel/signal name
  - Location (start bit) and size of the channel in a message
  - Signal type
  - Byte order (Intel/Motorola): Little-endian or big-endian
  - Scaling factor and units string
  - Range (minimum or maximum)
  - offset

# An Example CAN Database File




1 signal

data = 0x03E8 = $(1000)_d$

Database

| CAN message Name | Message ID | DLC | Scaling factor (2 bytes) | offset | Unit |
|---|---|---|---|---|---|
| Speed | 0x1E2 | 2 | 0.2 (resolution rpm/bit) | 100 | rpm |

⇒ raw data = 1000

actual data = raw data × scaling factor + offset

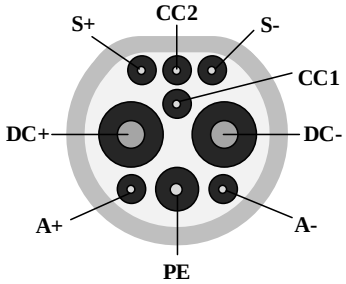
= 1000 × 0.2 + 100

= (200 + 100) rpm

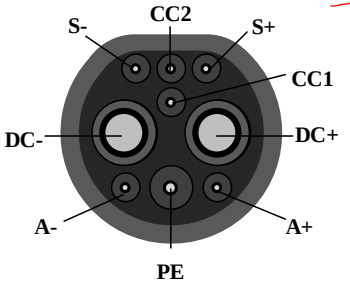
actual data = 300 rpm = speed value

# Recap: BEVC-DC001

- Input: 3-phase AC
- Output: Supports up to a maximum of 10kW for 48V system and 15 kW for 72V systems
- Communication is derived from IEC 61851-24B and GB/T 27930 → AIS 138 Part 2 (Annexure G)
- Uses CAN communication
- GB/T 20234.3 Connector with 5 m Cable




Charger Plug



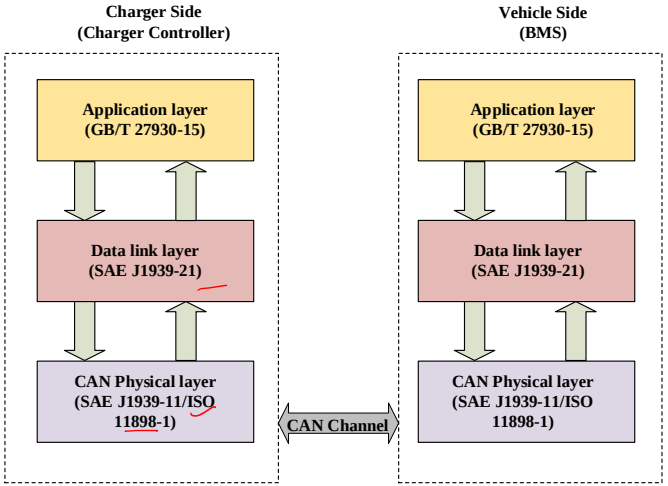

Vehicle inlet connector

**Introduced by Department of Heavy Industries, Government of India**

Source: https://heavyindustries.gov.in/sites/default/files/2023-09/Standardization%20of%20protocol.pdf

# Recap: Communication in DC001 (Derived from GB/T 27930)

- GB/T 27930-2015 defines rules for implementing the physical and data link layers based on SAE J1939 standard.
- Application layer of GB/T 27930-2015 defines representation of parameters and parameter group numbers.
- Speed – 250Kbps, uses CAN 2.0B
- Only two participants at a time – a charger and a BMS in the vehicle – with fixed addresses:
  - 86 (56Hex) for the charger
  - 244 (F4Hex) for the BMS
- No direct connections to other CAN systems in the vehicle, such as the power train CAN.
- CAN data frames are transmitted in the form of Protocol Data Unit (PDU) defined in GB/T 27930.




Source: A. V S, B. Y V, U. Sagar V and C. V, "Implementation and Test Bench Development of GB/T 27930-2015 Communication Protocol for Electric Vehicle Charger," in *IEEE Transactions on Industry Applications*, vol. 60, no. 5, pp. 7380-7388, Sept.-Oct. 2024, doi: 10.1109/TIA.2024.3425570.

# BEVC-DC001

- Physical layer include information like baud rate- 250kbps, CAN2.0B identifier (29 bit identifier), impedance of 120 ohm, etc.




- GB/T 27930-2015 specifies CAN data frame formats (message ID formats). CAN data frames are transmitted in the form of protocol data unit (PDU).
  - 3-bit priority field
  - Reserved bit
  - Data page bit
  - 8-bit- PGN → parameter group number
  - 8-bit destination address
  - 8-bit source address

29 bit Message ID
8 x 4 bit
29 bit + 3 bit
↳ avoided
4 bytes of message ID

Source: A. V S, B. Y V, U. Sagar V and C. V, "Implementation and Test Bench Development of GB/T 27930-2015 Communication Protocol for Electric Vehicle Charger," in *IEEE Transactions on Industry Applications*, vol. 60, no. 5, pp. 7380-7388, Sept.-Oct. 2024, doi: 10.1109/TIA.2024.3425570.

# BEVC-DC001

- All charging parameters are grouped into different parameter groups which are defined by parameter group number (PGN).
- Fixed address for **BMS = 244 (F4 Hex), EV charger = 86 (56 Hex)**
- The application layer of GB/T 27930-2015 is defined in the form of parameters and parameter group numbers.
- Both charger and EV are to be programmed to identify the data contained with the help of PGN of a message.
- Within each parameter group, there are signals with unique signal parameter number (SPN).

Message → Message ID → PGN → Parameter group → SPN

Source: A. V S, B. Y V, U. Sagar V and C. V, "Implementation and Test Bench Development of GB/T 27930-2015 Communication Protocol for Electric Vehicle Charger," in *IEEE Transactions on Industry Applications*, vol. 60, no. 5, pp. 7380-7388, Sept.-Oct. 2024, doi: 10.1109/TIA.2024.3425570.

# BEVC-DC001 Charging Stages and Messages

- Different charging stages: It has six stages of charging, and each stage has a set of messages to be transmitted and received.
- Six stages are
  - **Handshake stage**: charger handshake message (CHM), battery handshake message (BHM).
  - **Handshake recognition Stage**: Charger recognition message (CRM), Battery recognition (BRM) message.
  - **Parameter configuration stage**: Battery charge parameter (BCP) message, charger time synchronization message (CTS), charger max/min output parameter message (CML), charger output ready message (CRO), vehicle charge ready (BRO) message.
  - **Charging stage**: Battery charge requirement (BCL) message, Charger charge state (CCS) message, battery charge state 1 message (BSM), battery charge state 2 (BCS) message.
  - **Charging suspension stage**: Charger stopping command (CST) message, vehicle stopping command (BST) message.
  - **End-of-charging stage**: Charger statistic data (CSD) message, vehicle statistic data (BSD) message, vehicle receiving error, charger receiving error message
- Messages are as per AIS-138 part 2 (Annexure G) which is derived from GB/T 27930. However, in GB/T 27930message names and parameters are different. However, fundamentally structure is the same.

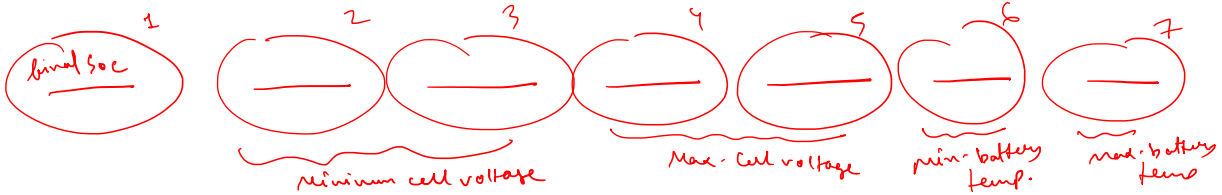
# BEVC-DC001

- The application layer in BEVC-DC001 is derived from GB/T 27930-2015 with few changes.
- Changes include addition of certain parameters some of them are (The details can be obtained from document released by MHI, GOI related to BEVC DC001 and AIS-138 part 2 (Annexure G)).
  - protocol version number (in charger handshake message (CHM)),
  - Battery type, battery manufacturer name and standard ASCII code, battery pack number, battery pack product code, battery pack charging times, battery pack property right mark, Vehicle Identification Number (VIN), etc in vehicle/battery recognition parameter message.
  - Charger number, charging station location code, recognition result, etc in charger recognition message.
  - Last charging end reason, Last charge status, Start state of charge (SoC) during last charging, end SoC during last charging, distance travelled since last charge.
  - List of fault codes

Source: A. V S, B. Y V, U. Sagar V and C. V, "Implementation and Test Bench Development of GB/T 27930-2015 Communication Protocol for Electric Vehicle Charger," in *IEEE Transactions on Industry Applications*, vol. 60, no. 5, pp. 7380-7388, Sept.-Oct. 2024, doi: 10.1109/TIA.2024.3425570.

# Example CAN Database file

| Message description | Message code | Source address | Destination address | PGN (Hex) | Priority | Data length /byte | Mandatory byte | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Starting position | Length /byte | Description |
| Vehicle statistic data | BSD | BMS | Charger | 001C00H | 6 | 7 | 1 | 1 | The final SOC |
| | | | | | | | 2 | 2 | Minimum cell voltage |
| | | | | | | | 4 | 2 | Maximum cell voltage |
| | | | | | | | 6 | 1 | Minimum battery temperature |
| | | | | | | | 7 | 1 | Maximum battery temperature |
| Battery charge requirement | BCL | BMS | Charger | 001000H | 6 | 5 | 1 | 2 | Voltage requirement |
| | | | | | | | 3 | 2 | Current requirement |
| | | | | | | | 5 | 1 | Charge mode |




The details can be obtained from AIS-138 part 2 (Annexure G) and GB/T 27930

Source for obtaining table for GB/T 27930 database information: Jiyong Chen, Yunjun Luo, Jianghui Jiang, Dian Lv "Analysis of Communication Protocol Standard for Conductive Charging of Electric Vehicles Based on GB/T 27930-2015"Academic Journal of Engineering and Technology Science, 2022, 5(4); doi: 10.25236/AJETS.2022.050402.

# Example CAN Database file

| Message description | Message code | Source address | Destination address | PGN (Hex) | Priority | Data length /byte | Mandatory byte | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Starting position | Length /byte | Description |
| Charger Max/min output parameter | CML | Charger | BMS | 000800H | 6 | 8 | 1 | 2 | Maximum output voltage |
| | | | | | | | 3 | 2 | Minimum output voltage |
| | | | | | | | 5 | 2 | Maximum output current |
| | | | | | | | 7 | 2 | Minimum output current |
| Charger recognition message | CRM | Charger | BMS | 000100H | 6 | 8 | 1 | 1 | Recognition result |
| | | | | | | | 2 | 4 | Charger number |
| | | | | | | | 6 | 3 | Charger/charge station location code |

The details can be obtained from AIS-138 part 2 (Annexure G) and GB/T 27930

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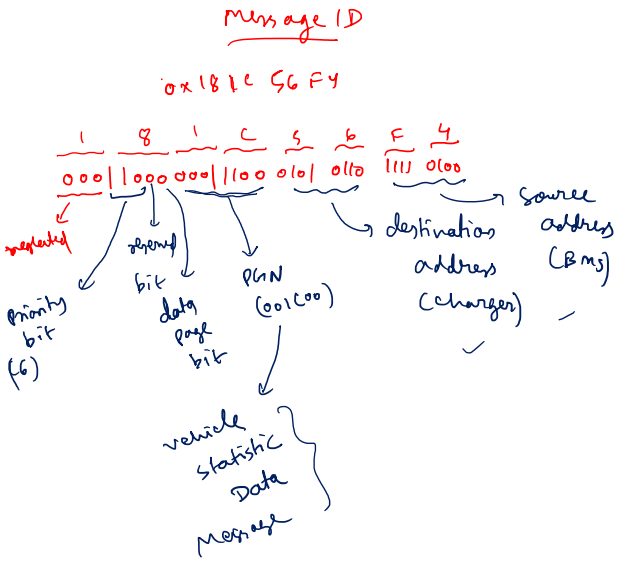
# An Example: Vehicle Statistic Data Message

| Start byte of bit | Length (byes) | Definition | Delivery option | Data resolution | Offset |
|---|---|---|---|---|---|
| 1 | 1 | The final SOC | Mandatory | 1% /bit | 0 |
| 2 | 2 | Minimum cell voltage | Mandatory | 0.01V/bit | 0 |
| 4 | 2 | Maximum cell voltage | Mandatory | 0.01V/bit | 0 |
| 6 | 1 | Minimum battery temperature | Mandatory | 1 deg C/bit | 0 |
| 7 | 1 | Maximum battery temperature | Mandatory | 1 deg C/bit | 0 |

PGN → 001C00 (Hex)

Source for obtaining table for GB/T 27930 database information: Jiyong Chen, Yunjun Luo, Jianghui Jiang, Dian Lv "Analysis of Communication Protocol Standard for Conductive Charging of Electric Vehicles Based on GB/T 27930-2015"Academic Journal of Engineering and Technology Science, 2022, 5(4); doi: 10.25236/AJETS.2022.050402.

# An Example: Vehicle Statistic Data Message




Message Data

46 012C 014A 0A3700 → (Hex.) (Raw data)

1st 2nd 3rd 4th 5th 6th 7th 8th (Byte order)

The final SOC = $(46)_H$ = $(70)_d$ = (raw data) $(70 \times 1\%/bit)$ (resolution) $+0$ (offset) = 70%

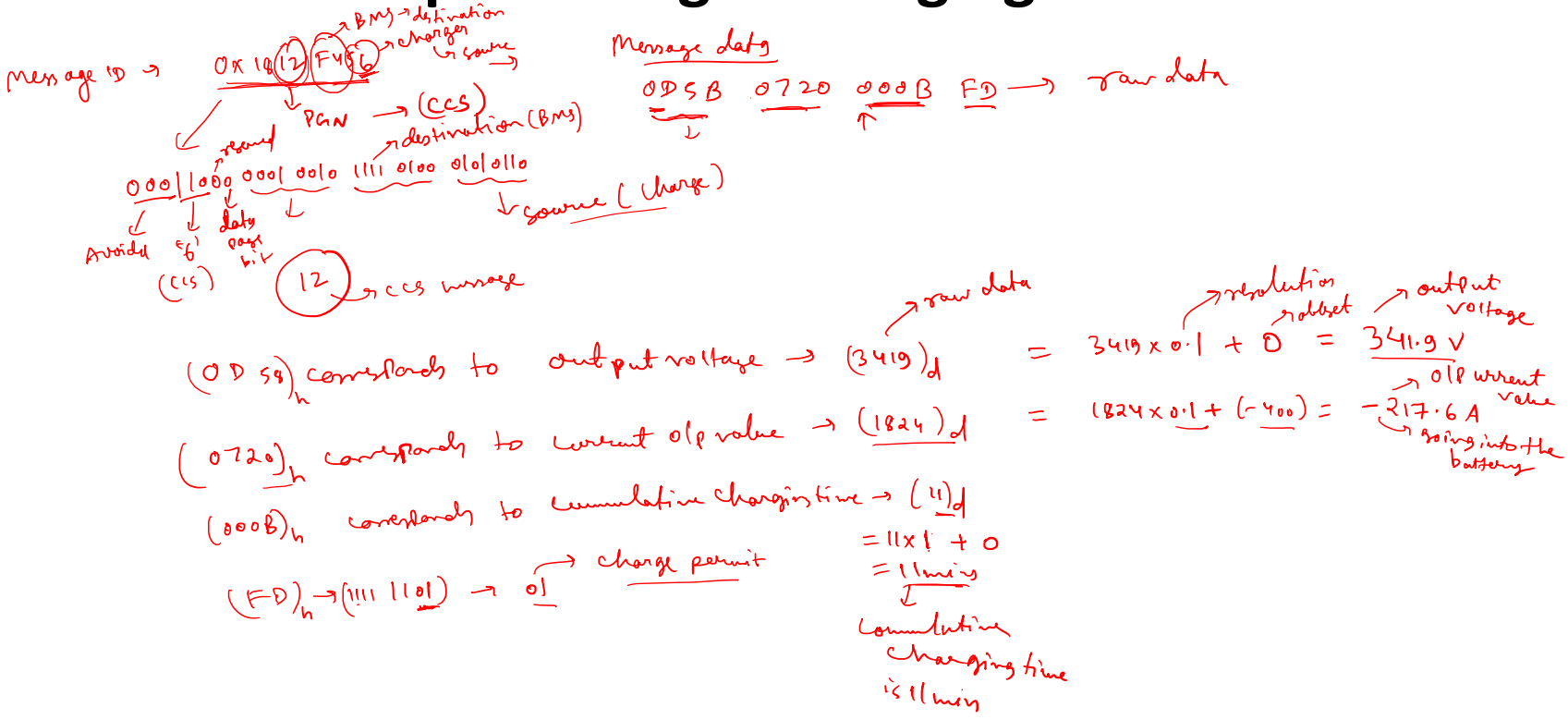
Minimum cell voltage = $(012C)_H$ = $(300)_d$ = $(300 \times 0.01V/bit) + 0$ = 3V

Maximum cell voltage = $(014A)_H$ = $(330)_d$ = $(330 \times 0.01V/bit) + 0$ = 3.3V

Minimum battery temperature = $(0A)_H$ = $(10)_d$ = $(10 \times 1°C/bit) + 0$ = 10°C

Maximum battery temperature = $(37)_H$ = $(55)_d$ = $(55 \times 1°C/bit) + 0$ = 55°C

# An Example: Charger Charging State



# Thank You